Course
« *Deep Learning*»

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2. Principal Component Analysis

Basic concepts

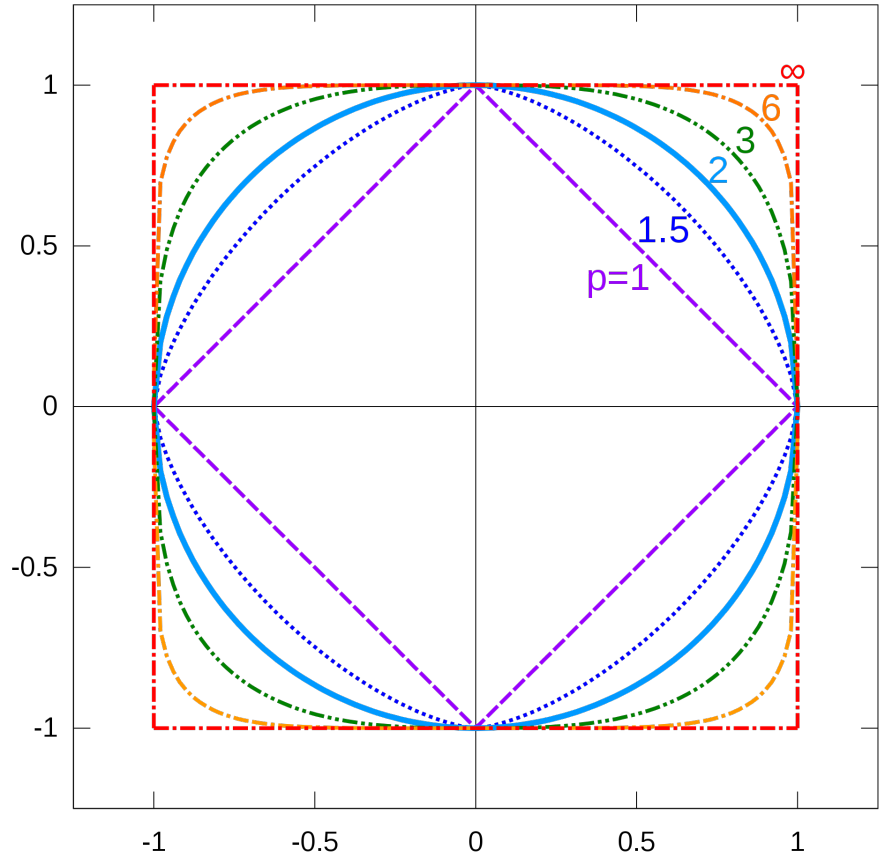
- **Distances**

$$\|\vec{x}\|_p = \left(|x_1|^p + \ldots + |x_n|^p\right)^{\frac{1}{p}}$$

$$\|\vec{x}\|_1 = |x_1| + \ldots + |x_n|$$

$$\|\vec{x}\|_2 = \sqrt{|x_1|^2 + \ldots + |x_n|^2}$$

$$\|\vec{x}\|_\infty = \max\left(|x_1|, \ldots, |x_n|\right)$$



L'ensemble des vecteurs de *p*-norme 1 dans $\mathbf{R}^2$ pour différentes valeurs de *p*.

And also, similarity measures…

2. Principal Component Analysis

Basic concepts

- **Variance:**

  Characterize the spread of the data in the space.

- The mean of the squared distances between data points x_i and their centroid *(c)*.

$$\frac{1}{n}\sum_{i=1}^{n} d(x_i, c)^2$$

2. Principal Component Analysis

Basic concepts

- *The tool* for any engineer who handles quantitative and multidimensional data.
- Fundamental method in multidimensional descriptive statistics.
- Simultaneous processing of any number of variables, all quantitative.
- Purely descriptive method: It does not assume any underlying model of a probabilistic type.

  $\Rightarrow$ When we consider a set of quantitative variables on which we would like to carry out a PCA, it is not necessary to assume that these variables follow normal distributions.

2. Principal Component Analysis

Basic concepts

- Simultaneously study a large number of quantitative variables (> 3).

  $\Rightarrow$ How to represent them graphically?

- **Solution :**
  - Go to a space of reduced dimension (for example 2) while distorting reality as little as possible.
  - Obtain the most relevant summary possible of the initial data.

- **This is one of the objectives of the PCA.**

2. Principal Component Analysis

Principle

- Data of dimension *p* => Data of dimension *q*
- Where *q* < *p*

Variables / features

Individuals

	X^1	$\cdots$	X^j	$\cdots$	X^p
1	x_1^1	$\cdots$	x_1^j	$\cdots$	x_1^p
$\vdots$	$\vdots$		$\vdots$		$\vdots$
i	x_i^1	$\cdots$	x_i^j	$\cdots$	x_i^p
$\vdots$	$\vdots$		$\vdots$		$\vdots$
n	x_n^1	$\cdots$	x_n^j	$\cdots$	x_n^p

Original data

Variables / features

Individuals

	U^1	$\cdots$	U^j	$\cdots$	U^q
1	u_1^1	$\cdots$	u_1^j	$\cdots$	u_1^q
$\vdots$	$\vdots$		$\vdots$		$\vdots$
i	u_i^1	$\cdots$	u_i^j	$\cdots$	u_i^q
$\vdots$	$\vdots$		$\vdots$		$\vdots$
n	u_n^1	$\cdots$	u_n^j	$\cdots$	u_n^q

Projected/transformed data

2. Principal Component Analysis

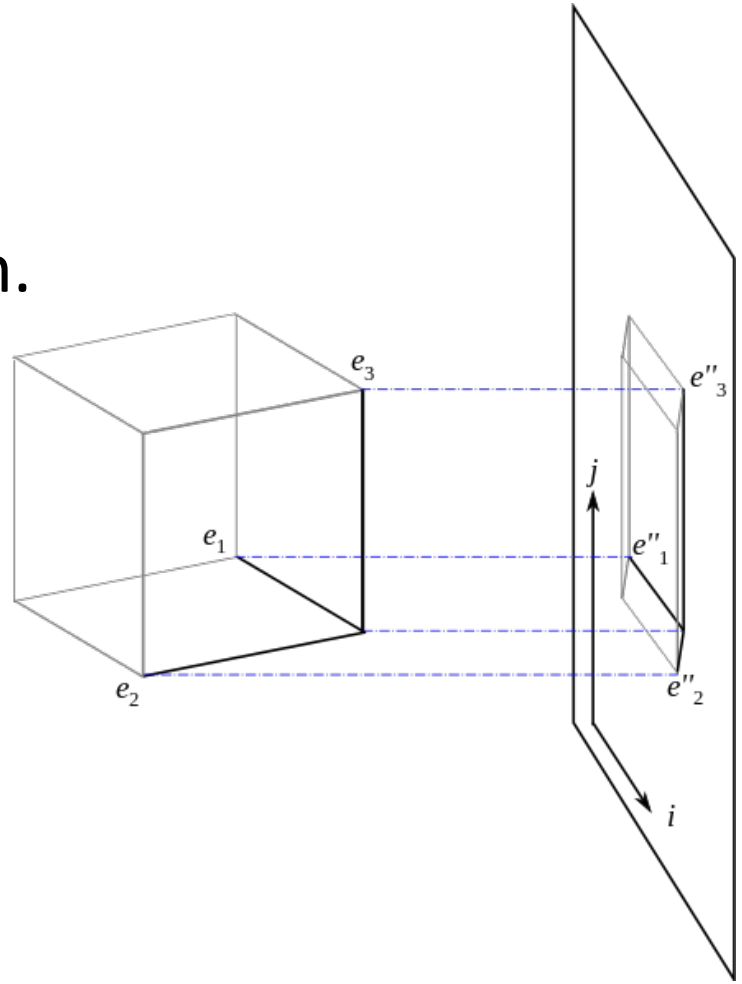
Principle

Projection => Information loss

=> Many different ways to perform.

Let’s consider the following example:





(Source : Wikipedia, projection orthogonale)

2. Principal Component Analysis

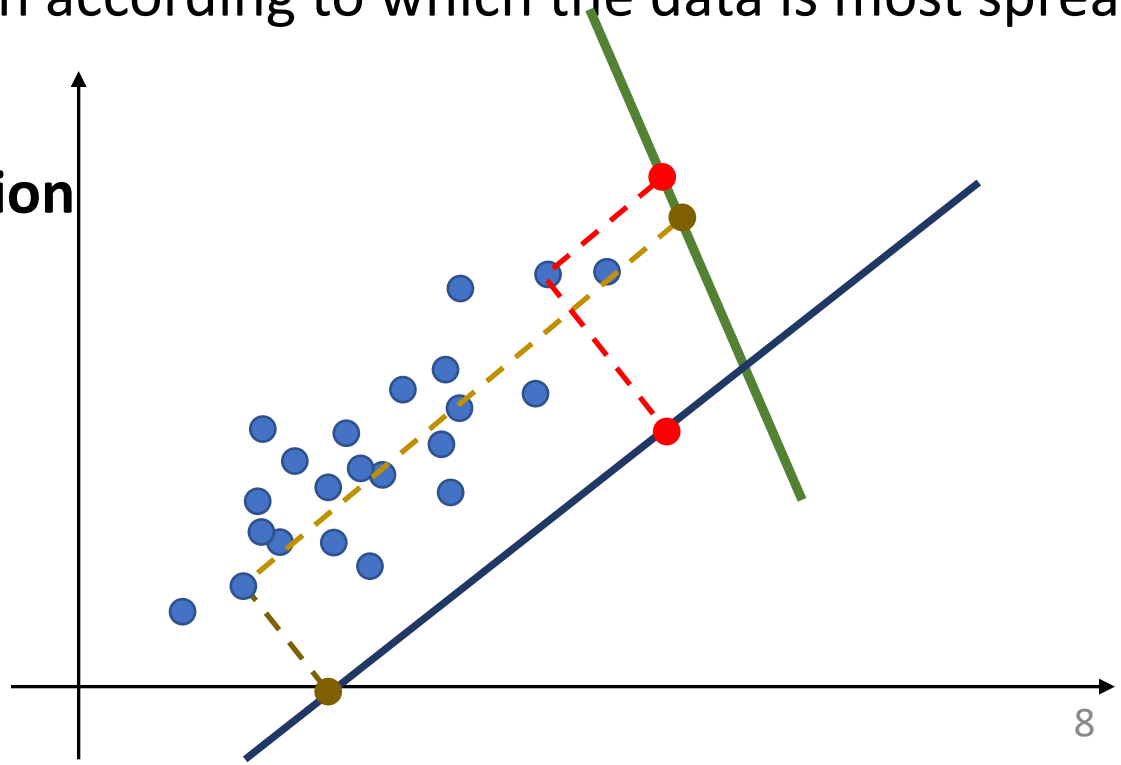
Principle

For a dataset => Find the projection that minimize information loss.

=> Maximize data dispersion (the variance).

Using the variance: Look for the projection according to which the data is most spread out.

Key principle of PCA: Choose the projection so as to maximize variance.



2. Principal Component Analysis

How does it work?

PCA therefore consists of searching, for a dimension $q < p$:

- The q factors maximizing the variance of the data when projected into the subspace of dimension q.
- By going from the initial dimension p to the reduced dimension q, we necessarily lose dispersion and then variance.
- The idea is to lose as little as possible by choosing properly the projection axes.

2. Principal Component Analysis

How does it work?

Find linear combinations of the initial variables:

$U^1 = v_1^1 X^1 + v_1^2 X^2 + \ldots + v_1^p X^p$.

$V^1 = (v_1^1, v_1^2, \ldots, v_1^p)$: Points are spread the most over this axis => ↑ variance

$U^2 = v_2^1 X^1 + v_2^2 X^2 + \ldots + v_2^p X^p$.

$V^2 = (v_2^1, v_2^2, \ldots, v_2^p)$: *Must spread points while* being not correlated to V^1.

Objective: V^2 brings a new information compared to V^1.

In practice, we add this additional constraint $\sum_{j=1}^{p} (v_i^j)^2 = 1$.

And so on…

2. Principal Component Analysis

How to do?

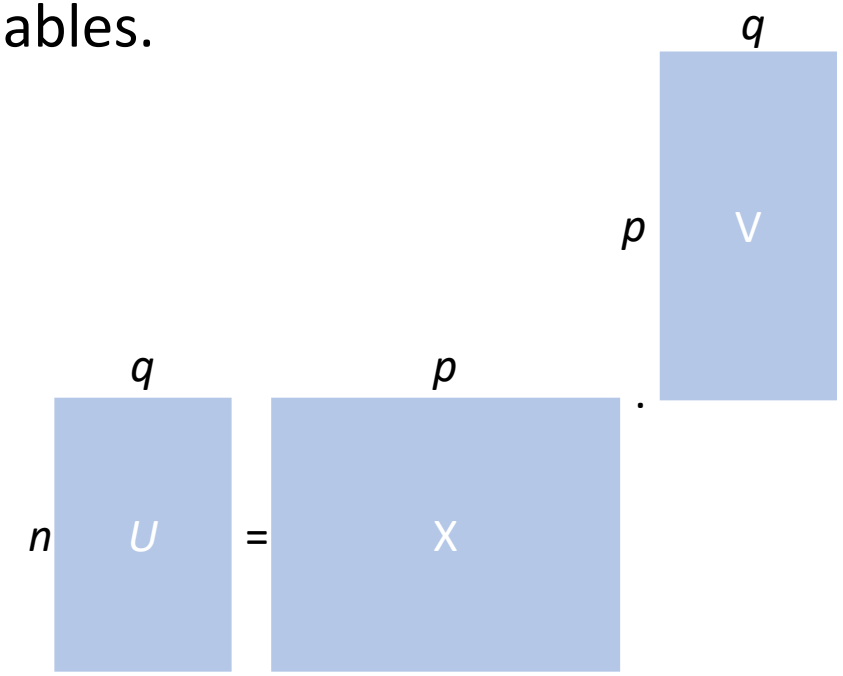
In order to reduce the dimension of the original space from *p* to *q*:

We have to find vectors $V^1, V^2, \dots, V^q$.

X: n x p — *n* individuals described using *p* variables.

V: p x q — *projection* matrix.

The new representation of the data: U = X . V

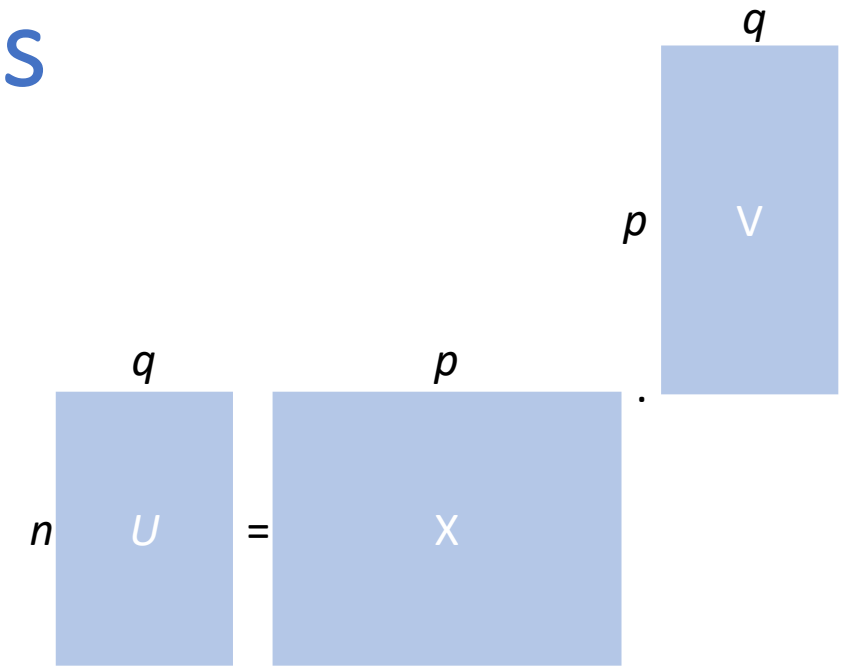


2. Principal Component Analysis

How to do?

Points in the new space: $U = X \,.\, V$

Reconstructing original points : $X' = U.\, V^t$



The objective: Find V such that X' is very close to X.

$$\min_{\mathbf{u},\mathbf{v}} J(\mathbf{u},\mathbf{v}) \qquad \text{avec} \qquad J(\mathbf{u},\mathbf{v}) = \sum_{i}^{n}\sum_{j}^{p}(x_{ij} - u_i v_j)^2$$

$$J(\mathbf{u},\mathbf{v}) = \|X - \mathbf{u}\mathbf{v}^\top\|_F^2$$

2. . Principal Component Analysis

It is then an optimization problem…

If we focus on *v* => Find a projection vector *(q=1)*

$$\min_{\mathbf{u},\mathbf{v}} J(\mathbf{u},\mathbf{v}) \quad \text{with} \quad J(\mathbf{u},\mathbf{v}) = \|X - \mathbf{u}\mathbf{v}^\top\|_F^2$$

In this case, **v** is the :

Eigen vector that corresponds to the largest eigen value of $\boldsymbol{X^T . X}$

and $\|\mathbf{v}\|_2 = 1$

It is the unique solution (Eckert & Young theorem 1936)

2. Principal Component Analysis

The algorithm

1. Center and reduce data *X (normalization step)*
2. Calculate the covariance matrix of the data: $X^T . X$
3. Calculate the eigenvalues and eigenvectors of the covariance matrix ($X^T . X$)
4. Order the pairs (eigenvalue: a_i, eigenvector: v_i) in descending order of eigenvalues
5. Keep the first q pairs in order to construct the matrix $V = [v_1\ v_2 \dots v_q]$.
6. Here: the eigen value a_i is equal to the inertia carried by the axis v_i.

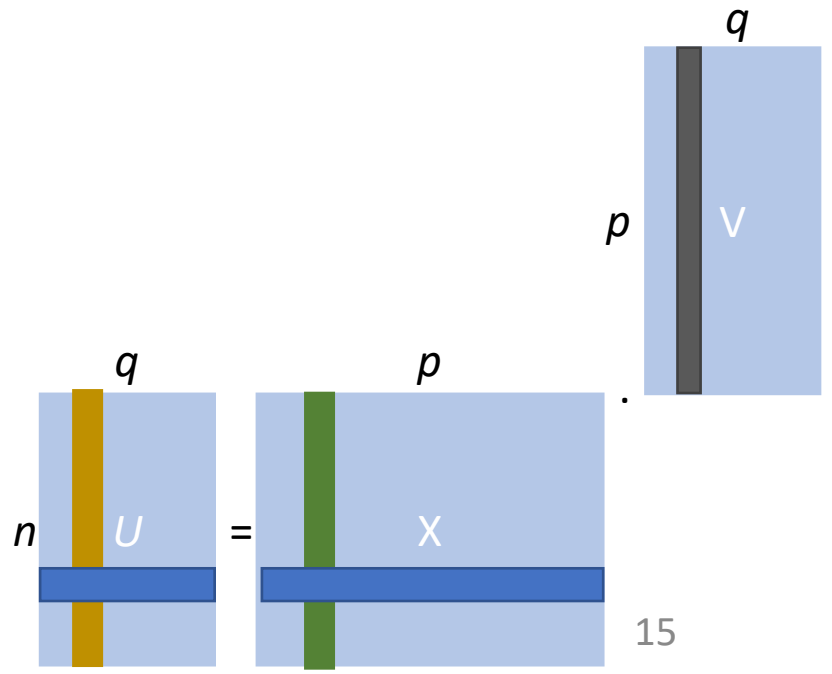
2. Principal Component Analysis

Three aspects to analyze

I. **The factors:** principal axes *(v, a_i),* i.e. eigen vectors and eigen values of the covariance matrix.

II. **The points:** the projection of the points in the new PCA space *(u)*

III. **The variables:** computing and analyzing the correlation between the initial variables and the principal components *(u).*



2. Principal Component Analysis

Question #1: Why center and scale the data?

- **Centering** => Subtracting the mean of each variable from each observation.
- **Scaling** => Dividing each centered observation by the standard deviation of the corresponding variable.

We perform **PCA with only centering** when the variables are homogeneous:

They have the same meaning, unit of measurement, and scale...

We perform **PCA with scaling** when the variables are heterogeneous.

2. Principal Component Analysis

Question #2: So, how do we determine the value of *q*?

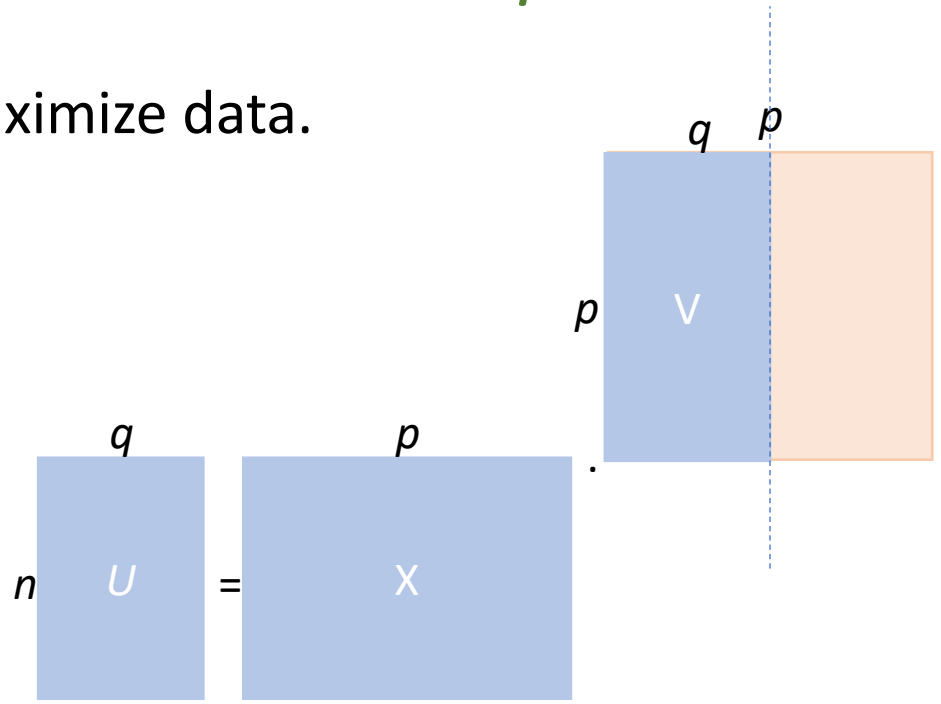
Recall that the primary goal of this approach is to maximize data. dispersion, and therefore the variance.

The most commonly used criterion is:

The explained variance ratio.

Choose q such that:

$$\frac{\sum_{i=1}^{q} a_i}{\sum_{j=1}^{p} a_j} > s$$



The proportion of the sum of retained eigenvalues relative to the sum of all eigenvalues should exceed a certain threshold (*s*).

2. Principal Component Analysis

Question #3: What are the objectives of PCA?

- By reducing dimensionality, PCA also allows us to study:

  Variability among individuals, meaning it helps identify differences and similarities between them.

  The data is thus represented in the new, reduced-dimensional space.

- PCA also enables the study of relationships between variables:

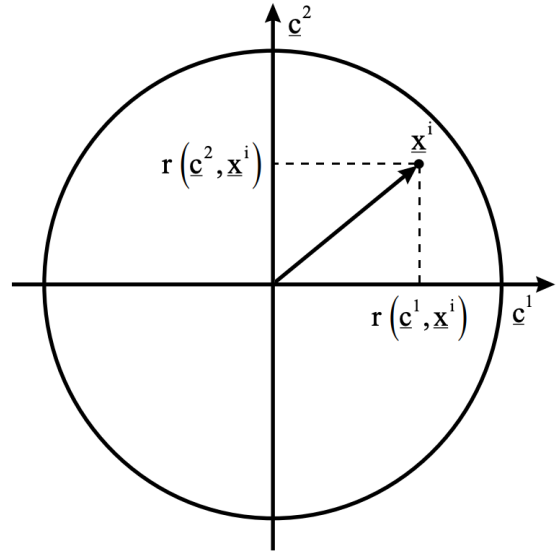
  Are there groups of highly correlated variables that can be combined into new synthetic variables?

2. Principal Component Analysis

Studying the relationships between variables

- Position yourself in a completely different space. The data table is structured as follows:
  - Each row corresponds to a 'variable.'
  - Each column represents an 'individual.'

- The goal is to analyze the linear correlation coefficients (r) between the initial variables (x^i) and the principal components (c^j)

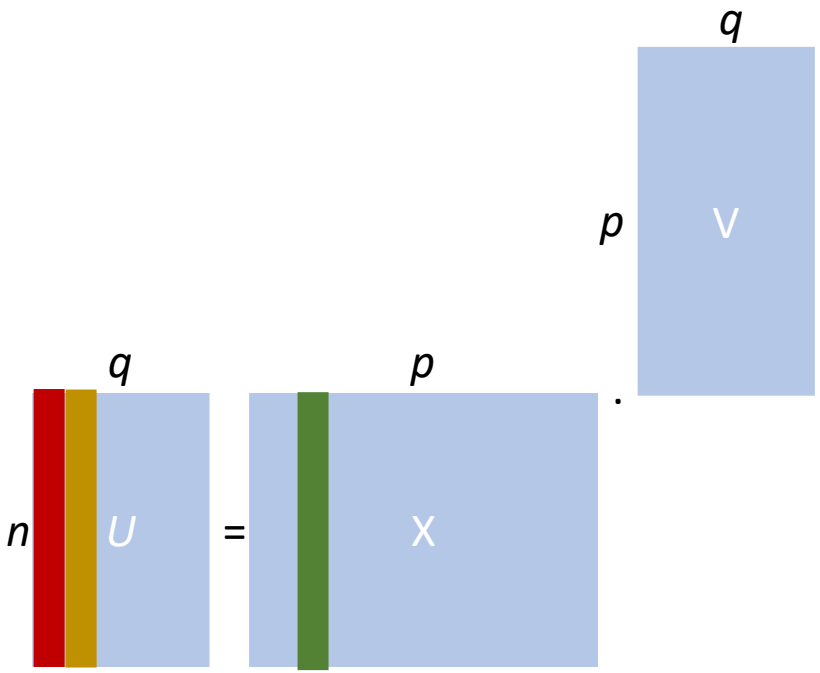


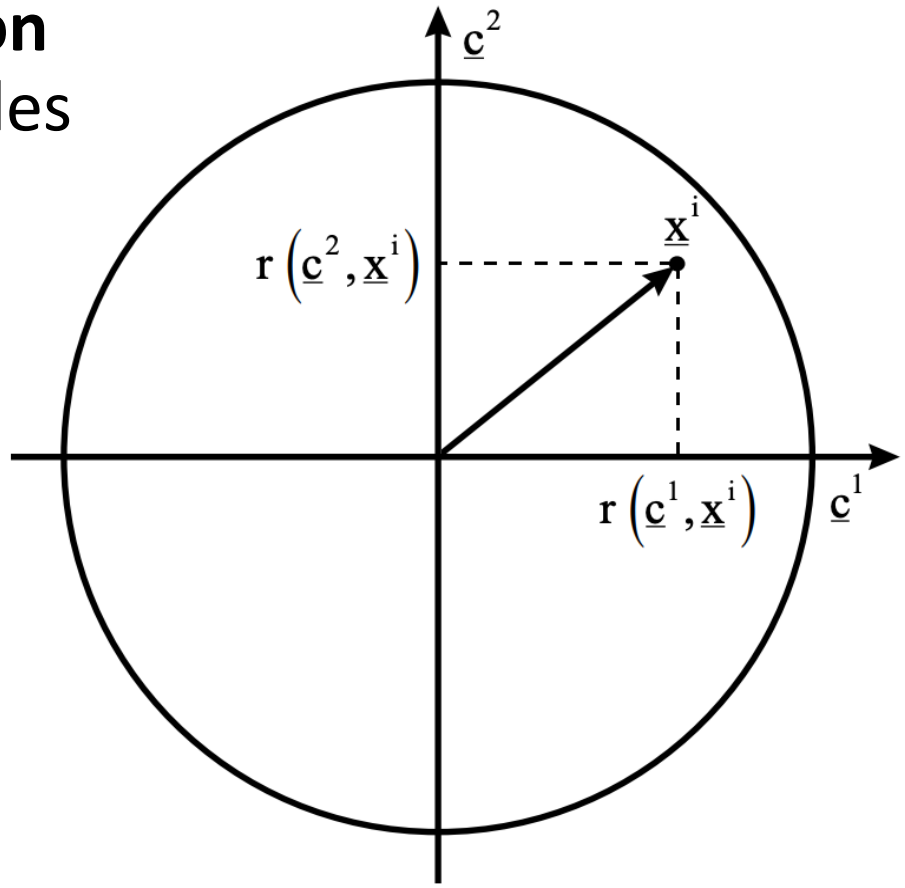
Correlation circle

2. Principal Component Analysis

Studying the relationships between variables

- The goal is to analyze the **linear correlation coefficients (r)** between the 'initial variables (x^i)' and the 'principal components (c^j).'



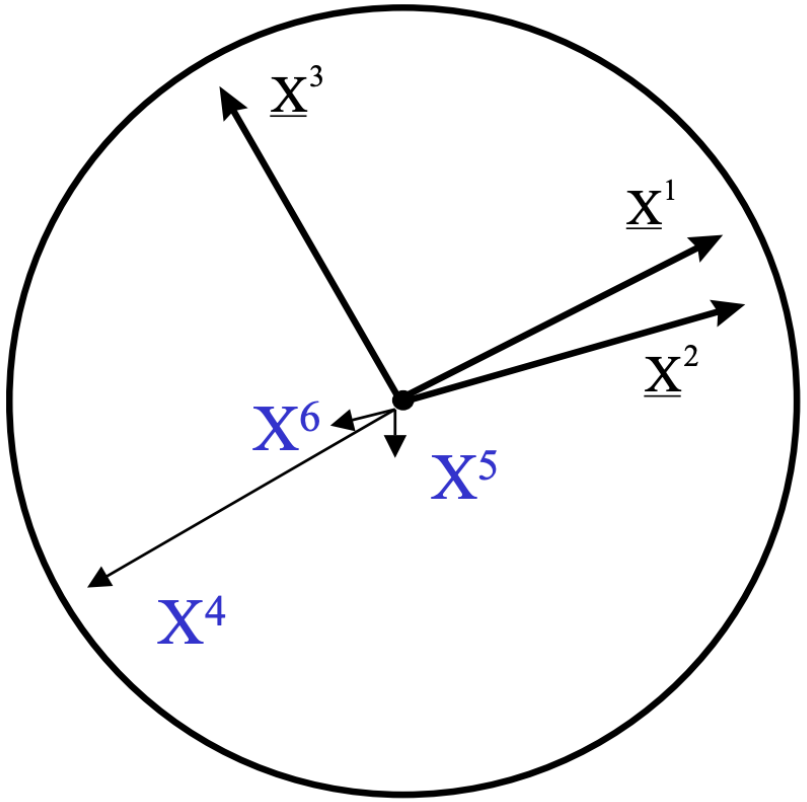


Correlation circle

2. Principal Component Analysis

Studying the relationships between variables

- X1 and X2 have a correlation close to 1.
- X1 and X3 have a correlation close to 0.



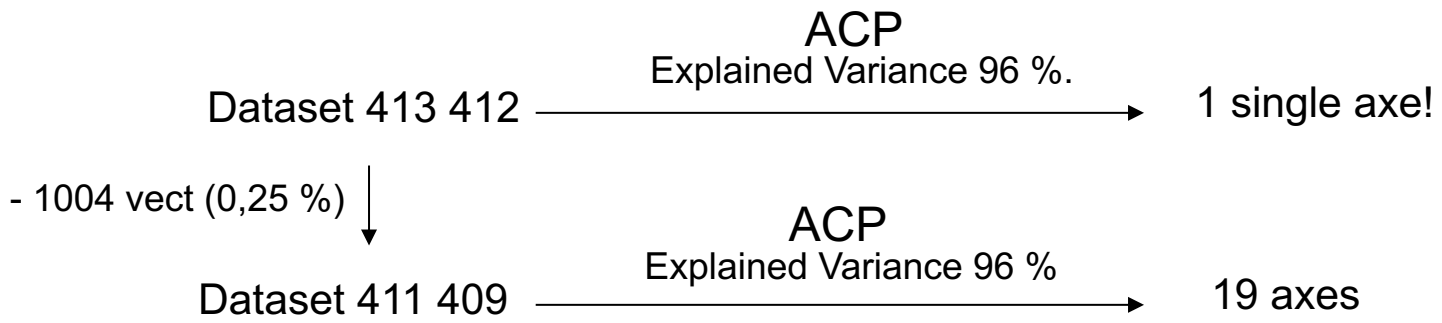
Correlation circle

2. Principal Component Analysis

The impact of outliers

- PCA is very sensitive to noise

Example :



2. Principal Component Analysis

A summary

PCA aims to:

- Study the **variability of individuals** (their similarities and differences).
- Analyze the **relationships between variables** (and if necessary, group-related variables into new synthetic variables to reduce the number of columns in our data).
- Find a representation space of **reduced dimension** while retaining the maximum amount of information:
  => This is a way to compress the data and eliminate noise and redundancies.
- Serve as an exploratory method.
  - Complex in its formulation but easy to implement (just a few lines of code)."

2. Principal Component Analysis

An example

Reference: https://www.math.univ-toulouse.fr/~baccini/zpedago/asdm.pdf

Student scores' analysis:

- 9 students (individuals)
- 4 subjects (variables)

	MATH	PHYS	FRAN	ANGL
jean	6.00	6.00	5.00	5.50
alan	8.00	8.00	8.00	8.00
anni	6.00	7.00	11.00	9.50
moni	14.50	14.50	15.50	15.00
didi	14.00	14.00	12.00	12.50
andr	11.00	10.00	5.50	7.00
pier	5.50	7.00	14.00	11.50
brig	13.00	12.50	8.50	9.50
evel	9.00	9.50	12.50	12.00

2. Principal Component Analysis

An example

	MATH	PHYS	FRAN	ANGL
jean	6.00	6.00	5.00	5.50
alan	8.00	8.00	8.00	8.00
anni	6.00	7.00	11.00	9.50
moni	14.50	14.50	15.50	15.00
didi	14.00	14.00	12.00	12.50
andr	11.00	10.00	5.50	7.00
pier	5.50	7.00	14.00	11.50
brig	13.00	12.50	8.50	9.50
evel	9.00	9.50	12.50	12.00

Elementary statistics:

Variable	Moyenne	Ecart-type	Minimum	Maximum
MATH	9.67	3.37	5.50	14.50
PHYS	9.83	2.99	6.00	14.50
FRAN	10.22	3.47	5.00	15.50
ANGL	10.06	2.81	5.50	15.00

Covariance matrix:

	MATH	PHYS	FRAN	ANGL
MATH	11.39	9.92	2.66	4.82
PHYS	9.92	8.94	4.12	5.48
FRAN	2.66	4.12	12.06	9.29
ANGL	4.82	5.48	9.29	7.91

2. Principal Component Analysis

An example

	MATH	PHYS	FRAN	ANGL
jean	6.00	6.00	5.00	5.50
alan	8.00	8.00	8.00	8.00
anni	6.00	7.00	11.00	9.50
moni	14.50	14.50	15.50	15.00
didi	14.00	14.00	12.00	12.50
andr	11.00	10.00	5.50	7.00
pier	5.50	7.00	14.00	11.50
brig	13.00	12.50	8.50	9.50
evel	9.00	9.50	12.50	12.00

Covariance matrix :

	MATH	PHYS	FRAN	ANGL
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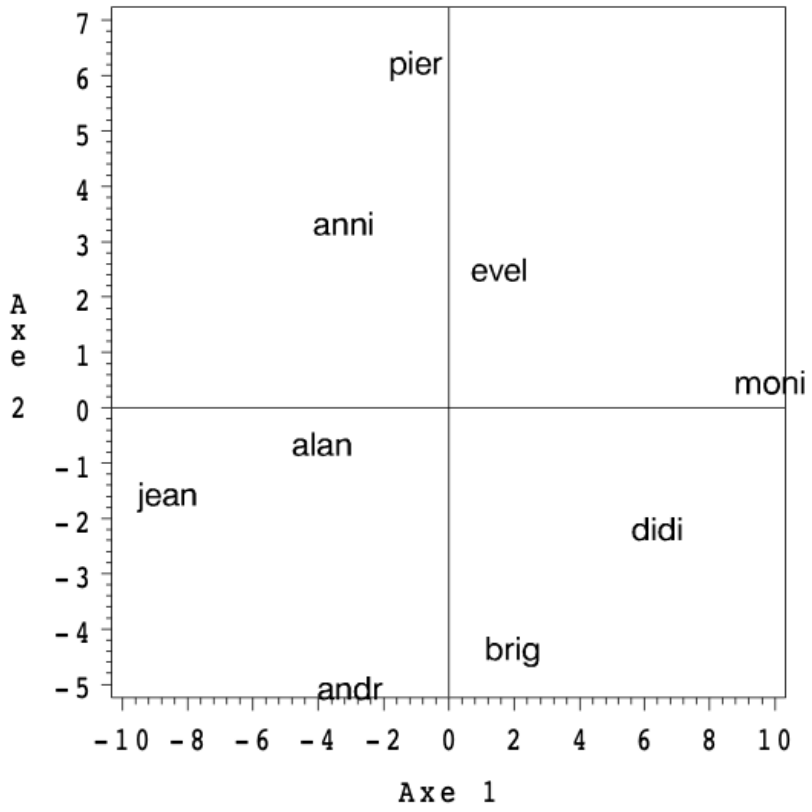
Eigen values and explained variance:

FACTEUR	VAL. PR.	PCT. VAR.	PCT. CUM.
1	28.23	0.70	0.70
2	12.03	0.30	1.00
3	0.03	0.00	1.00
4	0.01	0.00	1.00
	-----	----	
	40.30	1.00	

2. Principal Component Analysis

An example

Projection in a 2D space:



	MATH	PHYS	FRAN	ANGL
jean	6.00	6.00	5.00	5.50
alan	8.00	8.00	8.00	8.00
anni	6.00	7.00	11.00	9.50
moni	14.50	14.50	15.50	15.00
didi	14.00	14.00	12.00	12.50
andr	11.00	10.00	5.50	7.00
pier	5.50	7.00	14.00	11.50
brig	13.00	12.50	8.50	9.50
evel	9.00	9.50	12.50	12.00

	POIDS	FACT1	FACT2
jean	0.11	-8.61	-1.41
alan	0.11	-3.88	-0.50
anni	0.11	-3.21	3.47
moni	0.11	9.85	0.60
didi	0.11	6.41	-2.05
andr	0.11	-3.03	-4.92
pier	0.11	-1.03	6.38
brig	0.11	1.95	-4.20
evel	0.11	1.55	2.63

2. Principal Component Analysis

An example

	POIDS	FACT1	FACT2
jean	0.11	-8.61	-1.41
alan	0.11	-3.88	-0.50
anni	0.11	-3.21	3.47
moni	0.11	9.85	0.60
didi	0.11	6.41	-2.05
andr	0.11	-3.03	-4.92
pier	0.11	-1.03	6.38
brig	0.11	1.95	-4.20
evel	0.11	1.55	2.63

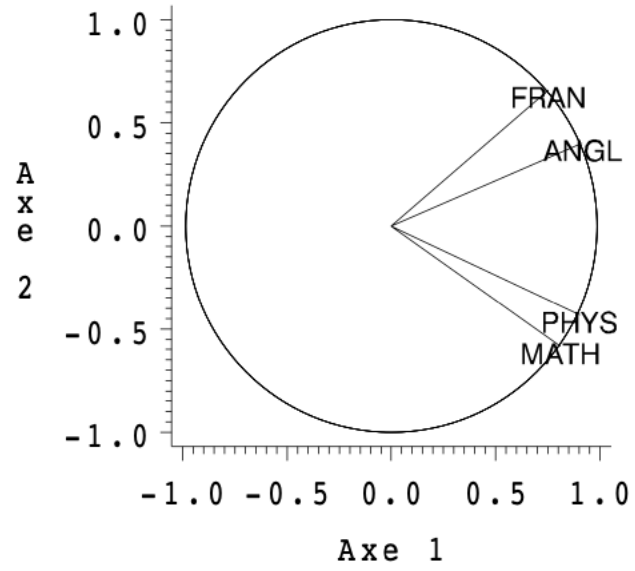
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didi	14.00	14.00	12.00	12.50
andr	11.00	10.00	5.50	7.00
pier	5.50	7.00	14.00	11.50
brig	13.00	12.50	8.50	9.50
evel	9.00	9.50	12.50	12.00

Results - variables

Correlations:

Corrélations variables-facteurs

FACTEURS -->	F1	F2	F3	F4
MATH	0.81	-0.58	0.01	-0.02
PHYS	0.90	-0.43	-0.03	0.02
FRAN	0.75	0.66	-0.02	-0.01
ANGL	0.91	0.40	0.05	0.01



2. Principal Component Analysis



2. Principal Component Analysis

Few references

- https://moodle.insa-rouen.fr/pluginfile.php/20201/mod_resource/content/5/slides/05_Slides_ACP_M8.pdf
- http://www.math.univ-toulouse.fr/~besse/pub/sdm2.pdf
- http://www.cs.cmu.edu/~elaw/papers/pca.pdf
- https://www.math.univ-toulouse.fr/~baccini/zpedago/asdm.pdf
- https://www.youtube.com/watch?v=aminVCp-olM